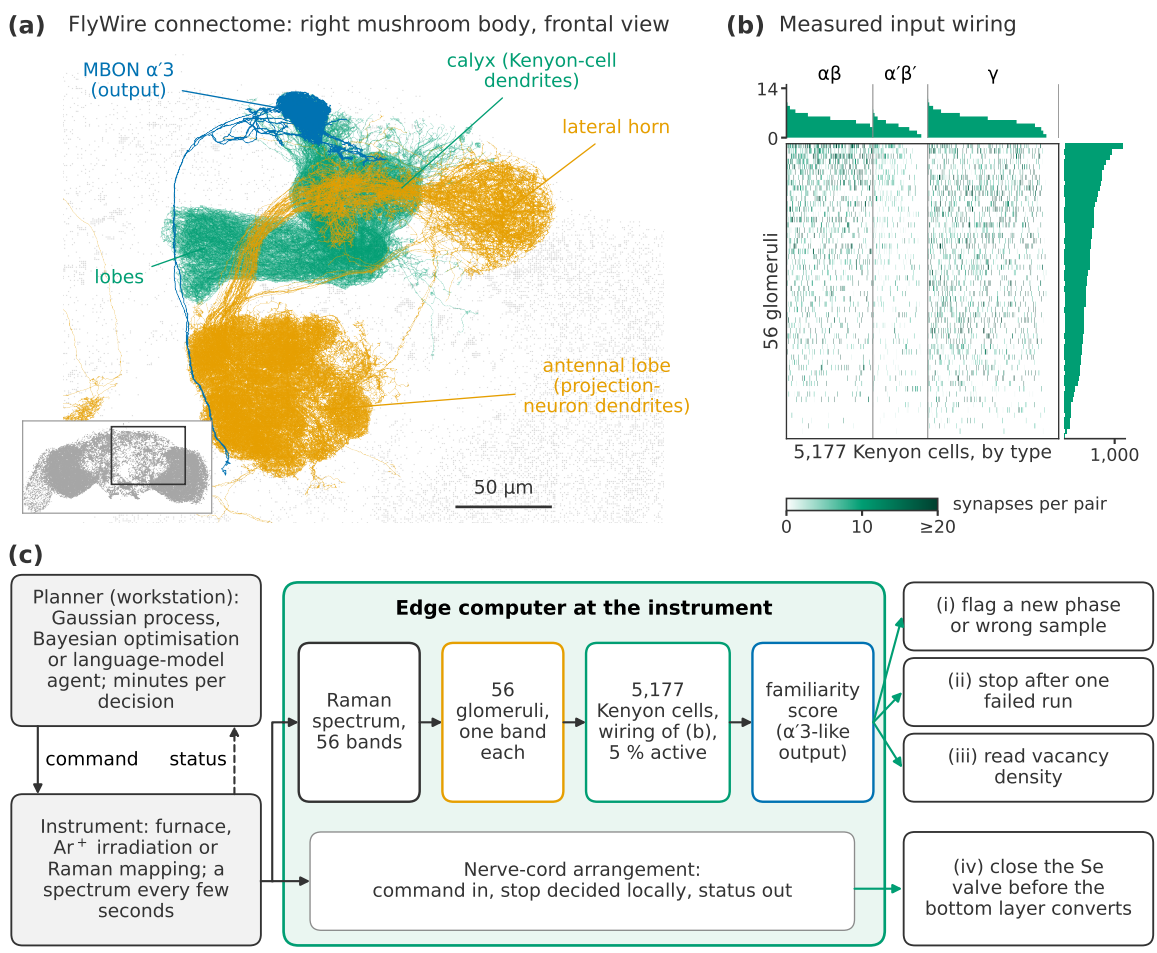

(a) FlyWire connectome: right mushroom body, frontal view
MBON α′3 (output)
calyx (Kenyon-cell dendrites)
lateral horn
lobes
antennal lobe (projection-neuron dendrites)
50 µm
(b) Measured input wiring
αβ
α′β′
γ
14
0
56 glomeruli
5,177 Kenyon cells, by type
1,000
synapses per pair
0
10
≥20
(c)
Planner (workstation): Gaussian process, Bayesian optimisation or language-model agent; minutes per decision
command
status
Instrument: furnace, Ar+ irradiation or Raman mapping; a spectrum every few seconds
Edge computer at the instrument
Raman spectrum, 56 bands
56 glomeruli, one band each
5,177 Kenyon cells, wiring of (b), 5 % active
familiarity score (α′3-like output)
Nerve-cord arrangement: command in, stop decided locally, status out
(i) flag a new phase or wrong sample
(ii) stop after one failed run
(iii) read vacancy density
(iv) close the Se valve before the bottom layer converts